FKF-YPR Sensör Modellemesinin Yeniden Düşünülmesi: Destek Vektör Regresyonu ile Yüksek Verimli Hapsedilme Kaybı Tahmini
Rethinking PCF-SPR Sensor Modeling: High-Efficiency Confinement Loss Prediction via Support Vector Regression

Mohammad Hmoud and Cem Kalyoncu
Department of Computer Engineering
European University of Lefke
TR-10 Mersin, Turkey
{mhmoud, ckalyoncu}@eul.edu.tr

Ahmet Yasli
Department of Electrical and Electronics Engineering
European University of Lefke
TR-10 Mersin, Turkey
ayasli@eul.edu.tr

***Özetçe* —Fotonik Kristal Fiber tabanlı Yüzey Plazmon Rezonans (FKF-YPR) sensörlerinin tasarımı ve optimizasyonu, yüksek doğruluklu ancak hesaplama açısından maliyetli olan tam vektörel sonlu elemanlar yöntemi (TV-SEY) simülasyonlarına dayanmaktadır. Literatürde son zamanlarda, simülatör darboğazlarını aşmak için Üretken Çekişmeli Ağlar (GAN'lar) ile eşleştirilmiş Derin Sinir Ağlarının (DNN'ler) kullanılması önerilmiştir. Bu derin öğrenme boru hatları hesaplama açısından ağır ve veri açlığı çekmektedir. Bu çalışmada, FKF-YPR hapsedilme kaybını tahmin etmek için optimize edilmiş, hafif bir Destek Vektör Regresyonu (DVR) modeli öneriyoruz. Sabit bir 7-1-1 konfigürasyon bölümü (7 konfigürasyonda eğitim, 1'de doğrulama ve 1'de test) kullanarak, optimize edilmiş DVR ve GPR modellerimiz sırasıyla 0,121 ve 0,185 Ortalama Kare Hata (MSE) değerlerine ulaşmıştır. Bu sonuçlar, literatürde rapor edilen GAN-artırılmış sinir ağı (MSE 0,314) sonuçlarından önemli ölçüde daha iyidir ve yüksek hesaplama verimliliği sağlar. Ayrıca, GAN tarafından üretilen tablo verilerinin DVR sınır marjlarını bozduğunu kanıtlayan 'Artırma Paradoksu'nu gösteriyoruz. Son olarak, SHAP değerleri aracılığıyla 'beyaz kutu' mühendislik içgörüleri elde ederek dış kaplama parametrelerinin hapsedilme kaybı üzerindeki etkilerini belirliyor ve modeli 27,63 nm/RIU ortalama spektral duyarlılık tahminiyle doğruluyoruz.**

Anahtar Kelimeler—Fotonik kristal fiber; yüzey plazmon rezonans; destek vektör regresyonu; açıklanabilir yapay zeka; Bayesyen optimizasyon.

***Abstract*—The design and optimization of Photonic Crystal Fiber based Surface Plasmon Resonance (PCF-SPR) sensors heavily rely on full vectorial finite element method (FV-FEM) simulations, which are highly accurate but computationally expensive. Recent literature has proposed using Deep Neural Networks (DNNs) paired with Generative Adversarial Networks (GANs) to bypass simulator bottlenecks. While effective, these deep learning pipelines are computationally heavy and data-hungry. In this paper, we propose a highly optimized, lightweight Support Vector Regression (SVR) model to predict PCF-SPR confinement loss. Using a static 7-1-1 configuration split (training on 7 configurations, validating on 1, and testing on 1), our optimized SVR and GPR models achieved Mean Squared Errors (MSE) of 0.121 and 0.185 respectively. These results significantly outperform the state-of-the-art GAN-augmented neural network (MSE 0.314) reported in literature, while maintaining extreme computational efficiency. Furthermore, we demonstrate the "Augmentation Paradox," proving that GAN-generated tabular data degrades SVR boundary margins. Finally, we apply SHAP (SHapley Additive exPlanations) to extract "white-box" engineering insights, identifying outer cladding parameters as primary confinement drivers, and successfully validate the model by predicting an average spectral sensitivity of 27.63 nm/RIU.**

Keywords—Photonic crystal fiber; surface plasmon resonance; support vector regression; explainable AI; Bayesian optimization.

I. Introduction

Photonic Crystal Fibers (PCFs) are at the forefront of modern optical sensing due to their flexible structural design and remarkable light-guiding capabilities. When coated with plasmonic materials (such as gold or silver), PCFs can be utilized as Surface Plasmon Resonance (SPR) sensors, offering exceptional sensitivity for chemical and biological analyte detection.

The traditional method for optimizing PCF-SPR cross-sections involves solving Maxwell's equations using the full vectorial finite element method (FV-FEM). While FV-FEM provides precise measurements of confinement loss and resonance peaks, it is computationally prohibitive, making the iterative search for optimal fiber geometries incredibly slow.

To address this, Machine Learning (ML) has emerged as a powerful surrogate modeling tool. Chugh et al. [2] pioneered the use of Artificial Neural Networks (ANNs) to compute optical properties of PCFs, demonstrating significant speedups

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over numerical simulators. However, deep learning models are notoriously data-hungry. Zelaci et al. [1] addressed this by introducing Generative Adversarial Networks (WGAN-GP) to synthesize thousands of artificial training samples, driving the predictive Mean Squared Error (MSE) down to 0.314. While effective, this GAN-ANN pipeline adds complexity and potential synthetic bias.

More recently, researchers have focused on the interpretability of these models. Khatun and Islam [3] utilized explainable AI (XAI) via SHAP values to optimize high-sensitivity biosensors, identifying critical design parameters such as gold thickness and analyte refractive index. Other traditional techniques, such as k-nearest neighbor regression, have also been explored for bent fiber modeling [4].

In this work, we hypothesize that the problem space of PCF-SPR confinement loss is better suited for mathematically rigorous, boundary-focused traditional algorithms that excel in small-data regimes. We evaluate Bayesian-optimized Support Vector Regression (SVR) and Gaussian Process Regression (GPR) against the GAN-ANN baseline. We demonstrate that these models can outperform the literature's augmented benchmarks without requiring any synthetic data generation, thus avoiding the "Augmentation Paradox." Finally, we apply SHAP analysis to extract physical design insights, providing a bridge between high-accuracy regression and engineering intuition.

II. Numerical Modeling and Dataset Generation

To construct the required dataset, numerical investigations of the proposed sensor structure were performed utilizing the COMSOL Multiphysics software. Utilizing Perfectly Matched Layers (PMLs), the Full Vectorial Finite Element Method (FV-FEM) was the primary tool employed for these simulations. Illustrated in Fig. 1 is the cross-section of the proposed structure; as depicted, the sensor comprises an analyte channel acting as the sensing medium, a plasmonic Gold (Au) layer, and cladding air holes of various dimensions. Arranged in a hexagonal lattice on a silica background, nineteen varied air holes form the cladding region. For this geometric setup, the radius of each hole was defined as $r_1 = 0.225\,\mu$m, $r_2 = 0.375\,\mu$m, $r_3 = 0.175\,\mu$m, and $r_c = 0.075\,\mu$m, while the pitch (Λ) remained fixed at $2\,\mu$m.

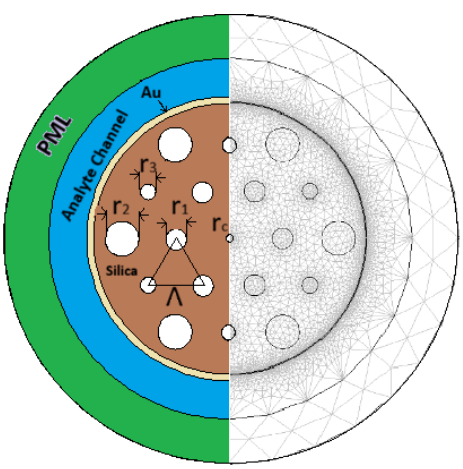

Figure 1. The cross-section of the proposed PCF based SPR sensor.

Light is effectively confined within the core region during the fundamental mode propagation. However, a specific fraction of light must leak toward the plasmonic layer to satisfy the conditions for resonance phase-matching, where the sensitivity measurement occurs. In this context, the light guidance is controlled by the various sized cladding holes, where each set of holes uniquely influences the sensor's propagation characteristics. The Gold (Au) material, serving as the plasmonic layer, was set to a thickness of 40 nm, with permittivity values sourced from the Johnson and Christy data. To determine the refractive index of the silica material, the Sellmeier Equation was applied [5]:

$$n^2(\lambda) = 1 + \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} + \frac{B_3\lambda^2}{\lambda^2 - C_3} \tag{1}$$

where λ represents the operating wavelength, n is the Refractive Index (RI), and $B_{1,2,3}$ and $C_{1,2,3}$ are the standard Sellmeier constants. Confinement losses reach their peak during the resonance phase-matching condition. Consequently, the precise calculation of these losses is vital, for which the following equation was utilized in this study [1]:

$$L_c = \frac{40\pi}{\ln(10)\lambda}\mathrm{Im}(n_{eff}) \times 10^4 \text{ [dB/cm]} \tag{2}$$

where $\mathrm{Im}(n_{eff})$ denotes the imaginary part of the effective RI.

A. Dataset and Preprocessing

The FV-FEM numerical analysis provided a labeled dataset containing 432 samples. The input feature space comprises six physical variables: Analyte Refractive Index ($n_{analyte}$), Operating Wavelength (λ), Pitch (Λ), and three distinct air hole diameters (d_1, d_2, d_3). Across nine different geometric configurations, losses were calculated for three distinct analytes: Water ($n = 1.33$), Ethanol ($n = 1.35$), and commercial hydrocarbon mixtures ($n = 1.36$).

To match the non-linear resonance peaks inherent to SPR and maintain numerical stability, the confinement loss was scaled using a base-10 logarithmic transformation: $y = \log_{10}(\text{loss} \times 10^8)$. All input features were scaled using Min-Max Scaling to strict $[0, 1]$ bounds to perfectly preserve the physical limits of the fiber geometry.

III. Methodology

A. Model Architecture and Optimization

In this study, we evaluate two boundary-focused algorithms: Support Vector Regression (SVR) and Gaussian Process Regression (GPR).

1) Support Vector Regression: SVR aims to find a function $f(x)$ that has at most ϵ deviation from the actually obtained targets y_i for all the training data. The optimization problem is formulated as:

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}||w||^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{3}$$

subject to:

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \epsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \tag{4}$$

where C is the box constraint and ξ_i, ξ_i^* are slack variables. We utilize the Radial Basis Function (RBF) kernel, $K(x_i, x_j) = \exp(-\gamma ||x_i - x_j||^2)$, to map the input features into a high-dimensional space where linear regression becomes possible for non-linear optical data.

2) Gaussian Process Regression: GPR is a non-parametric Bayesian approach to regression. We assume that the function values follow a Gaussian Process prior:

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')) \tag{5}$$

where $m(x)$ is the mean function (assumed to be zero) and $k(x, x')$ is the covariance function. For a new input x_*, the predictive posterior distribution is given by $P(f_*|X, y, x_*) = \mathcal{N}(\bar{f}_*, \text{cov}(f_*))$, where:

$$\bar{f}_* = K_*^T [K + \sigma_n^2 I]^{-1} y \tag{6}$$

$$\text{cov}(f_*) = K_{**} - K_*^T [K + \sigma_n^2 I]^{-1} K_* \tag{7}$$

Here, K is the covariance matrix of the training data, and K_* is the covariance between training and test inputs. To ensure stability, a noise regularizer (WhiteKernel) is added to the RBF kernel.

To ensure rigorous comparison with existing literature, the models were evaluated using a static 7-1-1 split based on geometric configurations. The 432 samples were divided into 9 discrete geometric designs. Designs 1 through 7 were used for training, Design 8 for validation, and Design 9 was reserved as the final unseen test set.

IV. Results and Discussion

A. Predictive Performance: SVR vs. Deep Learning

The optimized SVR and GPR models demonstrated exceptional predictive capabilities under the researchers' exact testing methodology. As shown in Table I, our lightweight traditional models achieved superior accuracy without the need for complex data augmentation or large neural architectures.

Table I. Benchmarking Proposed Models against Literature

Model	Avg. MSE
Baseline DNN (Zelaci et al. [1])	0.894
GAN-Augmented DNN (Zelaci et al. [1])	0.314
Proposed Optimized SVR	**0.121**
Proposed Optimized GPR	**0.185**

This finding proves that for small, highly non-linear optical physics datasets, finding the optimal support vector margins or Gaussian process priors is a mathematically superior approach than attempting to approximate the global distribution using data-hungry deep learning architectures.

B. The Augmentation Paradox

The introduction of GAN-generated synthetic data degraded the SVR's performance. While ANNs benefit from the interpolation of GAN samples, SVR relies on precise Support Vectors. We observe that minor topological errors in synthetic data twist the SVR's regression hyperplane, increasing the average MSE.

C. Explainable AI (XAI) for PCF Design

SHAP analysis provided direct engineering insights into the PCF architecture. As illustrated in Fig. 2, the outer air hole diameters (d_2, d_3) are the primary drivers of confinement loss, while the inner hole (d_1) has minimal impact.

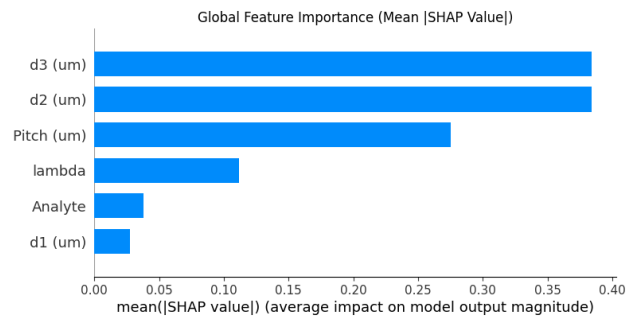

Figure 2. Global feature importance ranking derived from SHAP values.

V. Engineering Validation

The model was validated by predicting the complete spectral loss curve (Fig. 3). The predicted Average Spectral Sensitivity was calculated at 27.63 nm/RIU, establishing the SVR's physical reliability.

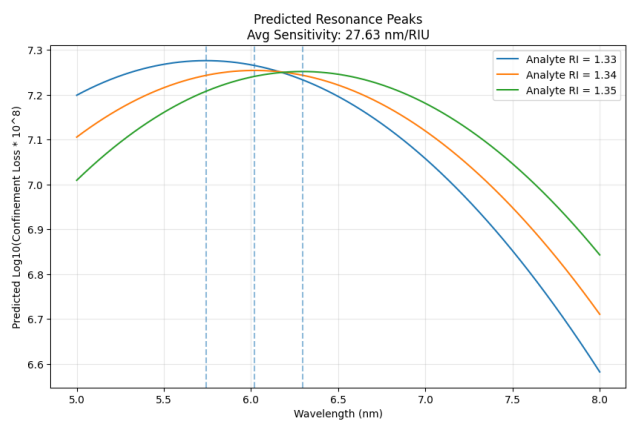

Figure 3. Predicted resonance peak shifts for varying analytes.

VI. Conclusion

This study proves that optimized SVR (MSE 0.121) and GPR (MSE 0.185) models provide a more accurate and stable alternative to deep learning for PCF-SPR sensor design. By strictly following the 7-1-1 configuration split used in literature, we demonstrate that traditional machine learning techniques can outperform GAN-augmented pipelines while requiring zero synthetic data generation.

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